Real Estate Price Trend Prediction

Multi-agent system with a supervisor and specialist workers

Project plan | October 13, 2026 | Version 1

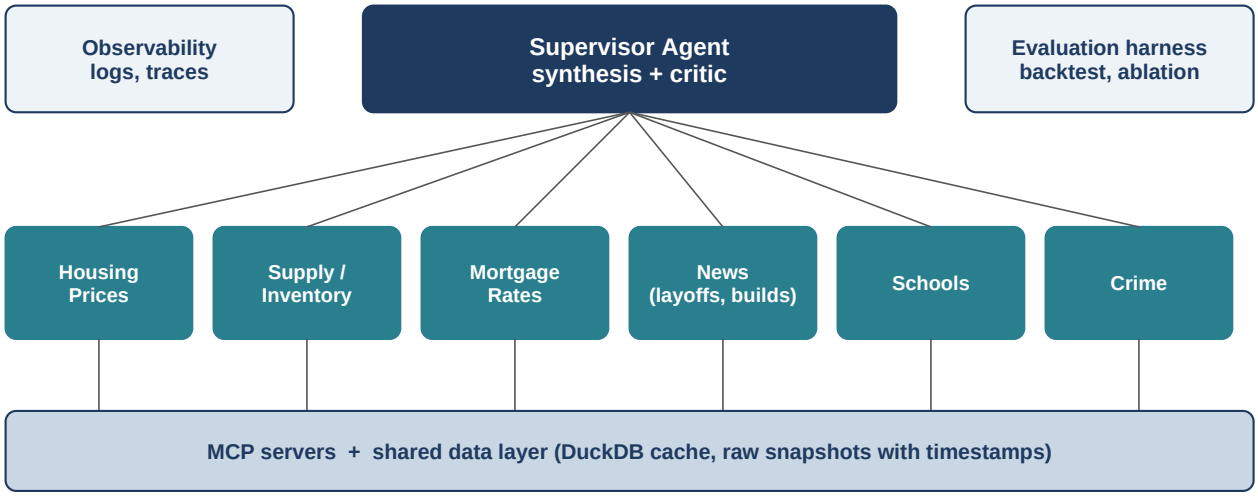
Goal. Build an autonomous system that produces a directional outlook (up, flat, down) with stated uncertainty for housing prices in a city. A supervisor agent combines structured reports from specialist worker agents and explains its conclusion with evidence. The system is built one component at a time and is validated by backtesting before more signals are added.

1. Design principles

- **Code for data, LLM for judgment.** Fetching data and computing trends is deterministic and should be plain code exposed as tools. The LLM is used only where judgment is needed, such as reading news or interpreting what a rate move means.
- **Every worker returns the same structured output** so the supervisor compares like with like and can weight by confidence and freshness instead of parsing free text.
- **Directional outlook, not a point price.** A fluent narrative is not a validated forecast. Conclusions carry stated uncertainty and link back to evidence.
- **Prove it before expanding.** No new signal is added until the backtest shows a gap it could fill.
- **Fail gracefully.** A missing or stale worker lowers confidence and is disclosed. It never fails silently.

2. Architecture

A supervisor delegates to specialist workers. Each worker uses narrow MCP tools backed by a shared data layer. Observability and evaluation sit alongside the supervisor from the start.



External sources: Zillow Research, Redfin, FRED, FBI CDE, NCES, news / WARN / Census permits

MCP structure. One MCP server per data domain (housing data, rates, crime, schools, news), each exposing narrow tools such as `get_zhvi_trend(location, years)`. A worker is then a prompt plus access to its server. This keeps tools reusable and lets a data source be swapped without touching agent logic. Before building a server, check whether one already exists (for example for FRED).

3. Worker agents

Six workers: the five from the original outline plus the supply and inventory agent. The last column shows how much of each worker needs an LLM.

Worker	Purpose	Primary data sources	Code vs. LLM
Housing prices	Price level and trend (1, 3, 5 year) for the location	Zillow Research CSVs (ZHVI, ZORI), Redfin Data Center, FHFA HPI, S&P Case-Shiller	Mostly code. Thin LLM layer to summarize.
Supply / inventory (priority)	Leading indicators: active listings, months of supply, new listings, days on market, share of listings with price cuts	Redfin Data Center market tracker	Mostly code. Thin LLM layer.
Mortgage rates	Current and recent rates and their likely effect on prices and affordability	FRED API, series MORTGAGE30US (Freddie Mac weekly)	Tool call plus LLM interpretation.
News	Layoffs and new construction as desirability signals	News search API or GDELT, WARN Act notices, Census Building Permits Survey, city permit portals	Needs the LLM. Noisiest worker.
Schools	School ratings and historical trends as a family desirability signal	NCES, state education department data. GreatSchools API only if licensing allows.	Tool plus summary.
Crime	Crime rate trends as a safety signal, with data-quality flags	FBI Crime Data Explorer API, city open-data portals (Socrata)	Tool plus summary. Must flag coverage gaps.

Note on Zillow and Redfin. Neither offers a public API for price data, and scraping breaks often and violates their terms. The downloadable research files above are structured, free, and monthly, so the housing agent is mostly download, parse, and compute trends.

4. Priority additions beyond the original outline

Three additions are in scope for version 1. The first improves the forecast. The other two let us find out whether any of it works.

Addition	What it is	Why it is a priority
Supply / inventory agent	Worker on Redfin market data covering listings, months of supply, days on market, price-cut share.	Often the best leading indicator of price direction, because it shifts before prices do. Cheap to add since the data source is shared with the housing agent.
Data layer with caching	DuckDB or SQLite store of raw pulls with timestamps, one table per source.	Sources update monthly or quarterly, so re-fetching each run is wasteful. Timestamped snapshots also give reproducibility and make historical backtests possible.
Evaluation harness	Backtest: run the system as of a past date using only data available then, and compare against the following 6 to 12 months. Ablation: drop each worker and measure the change.	Shows whether the system beats the plain Zillow trend line and which agents earn their place. Mortgage rates and price momentum will likely dominate, and this will confirm it.

5. Shared foundations

Two decisions come before any worker is built, because every worker depends on them.

5.1 Canonical location key

"City" is ambiguous. School districts, police jurisdictions, ZIP codes, and metros do not line up. Define one location object that every worker accepts, for example a city name and state, its list of ZIPs, its county FIPS code, and its metro (CBSA) code. Each worker maps this to its own source's geography and reports how closely the match fits.

5.2 Worker output contract

Every worker returns this structure. The supervisor never reads free text from a worker.

```
{
  "worker": "supply_inventory",
  "location": {"city": "...", "state": "..", "zips": [...], "fips": "...", "cbsa": "..."},
  "signal": "supply",
  "direction": "up | flat | down",
  "confidence": 0.0-1.0,
  "key_metrics": {"months_of_supply": 3.4, "price_cut_share": 0.21, ...},
  "data_as_of": "2026-09-30",
  "geography_match": "exact | approximate",
  "caveats": ["..."],
  "sources": [{"name": "...", "url": "...", "retrieved": "..."}]
}
```

6. Supervisor behavior

- **Validation (critic) step.** Before synthesizing, check for stale data, conflicting signals, geography mismatches, and missing workers. Decide whether to proceed, retry a worker, or lower confidence.
- **Graceful degradation.** If a worker is unavailable, say so explicitly (for example, "crime data unavailable, conclusion based on five of six signals").

- **Weighted synthesis.** Weight signals by confidence and freshness. Start with transparent weights that the evaluation harness can tune, not weights chosen by the LLM.
- **Output.** A directional outlook, an uncertainty statement, the top drivers, dissenting signals, and evidence links for every claim.
- **Observability.** Log every worker input, output, and tool call, and the supervisor's reasoning, so any odd conclusion can be traced to the signal that caused it.

7. Build phases

Each phase ends with something that runs. The integrated pipeline exists by the end of Phase 3, and everything after that is gated by evaluation results.

Phase	Work	Done when
0. Foundations	Choose agent framework. Define location key and output schema. Project skeleton. Data layer with caching (DuckDB). Structured logging.	A fake worker can write a schema-valid report and a cached pull can be replayed.
1. Mortgage rates	FRED tool (MCP server) and rate worker. Easiest end-to-end path.	Worker returns a valid report for any location, with freshness and sources.
2. Housing + supply	Zillow Research and Redfin loaders. Housing price worker and supply/inventory worker.	Both workers pass schema checks on at least three test cities.
3. Supervisor v1	Minimal supervisor combining the three workers so far, with critic step and failure handling.	End-to-end run produces an outlook with evidence for a test city.
4. Evaluation harness	Backtest runner using timestamped snapshots, baseline (plain Zillow trend), ablation by worker.	We can state whether the system beats the baseline and by how much.
5. Add signals	Crime, then schools, then news (noisiest last). After each, rerun the backtest and ablation.	Each added worker is kept only if it improves results or adds needed explanation.
6. Usability	Comparison mode across cities or ZIPs, structured report, scheduled monthly refreshes that track how the outlook changes.	A monthly run produces a report and flags outlook changes since last run.

8. Risks and mitigations

Risk	Mitigation
Zillow/Redfin data access or terms issues	Use published research files and Data Center downloads. No scraping.
Geography mismatch across sources	Canonical location key. Each worker reports match quality and the supervisor lowers confidence on approximate matches.
Confident narrative without predictive value	Directional outlook with uncertainty. Backtest against a naive baseline before trusting output.
Noisy or biased news signal	Prefer WARN notices and permit counts over headlines. Build the news worker last.
Crime data inconsistency across jurisdictions	Flag reporting changes and coverage gaps. Compare trends within a location over time, not across cities.
Stale or missing data	Freshness field in every report. Critic step and graceful degradation in the supervisor.
Scope creep	Backlog below is pulled in only when the backtest shows a gap.

9. Backlog (not in version 1)

Candidate	Value	Notes
Affordability and income	Price-to-income ratio, payment burden	Census ACS income data. Adds context to the rate signal.
Employment and economy	Unemployment and job growth by metro	BLS data. Pairs with the news worker as the measured aggregate.
Demographics and migration	Population change vs. housing supply growth	Census population estimates.
Rent and investor signal	Rent-to-price ratio, rent trend	Zillow ZORI is already in the housing loader.
Climate and insurance risk	Flood, wildfire, insurance cost exposure	Increasingly material in some regions.
Local policy	Zoning, property tax, rent control	Hard to automate. Last.

10. Open decisions

- **Agent framework:** Claude Agent SDK, LangGraph, or another. This affects how the supervisor and workers are laid out.
- **Starting geography:** two or three test cities with different market conditions, used throughout development and backtesting.
- **Backtest horizon and metric:** for example, directional accuracy over 6 and 12 months against a plain trend baseline.
- **Observability tooling:** structured JSON logs to start, or a tool such as Langfuse.
- **Language and storage:** Python and DuckDB assumed unless there is a reason to differ.